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Cerebral protection of purified *Herba Leonuri* extract on middle cerebral artery occluded rats

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A B S T R A C T

Aim of study: Oxidative stress is involved in stroke. In particular, Chinese Herbal Medicine with antioxidant properties is believed to have potential therapeutic effect. In this study, neuroprotective effects of purified *Herba Leonuri* (pHL) were evaluated in *Wistar* rats undergone middle cerebral artery occlusion (MCAO).
Materials and methods: The rats were treated with their respective treatments for 2 weeks prior to the MCAO, continually treated for another 7 days after MCAO. During the post-surgery treatment period, neurological deficit score was measured. At the end of treatment, animals were sacrificed and samples were collected for analysis of infarct volume, apoptosis and antioxidant analysis.
Results: Under the treatment of pHL, the infarct volume was reduced significantly from 20.75 ± 0.03% to 15.19 ± 0.02% ($p < 0.05$). The neurological impairment was alleviated to 1.82 as compared to vehicle (2.43). Plasma antioxidant concentration was increased from 0.31 ± 0.03 mM to 0.42 ± 0.05 mM ($p < 0.05$). DNA oxidative damage was reduced to 1.19 ± 0.03 in stroke pHL treated group ($p < 0.05$ as compared to vehicle group, 1.78 ± 0.03). pHL could reduce the level of apoptosis and also the pro-apoptotic proteins, but increase the level of anti-apoptotic proteins.
Conclusion: pHL is believed to have promising therapeutic effect for stroke treatment through antioxidant mechanisms.

1. Introduction

Ischemic stroke is caused by a reduction or complete blockade of blood flow, resulting in the deficiency of glucose and oxygen supply to the territory of the affected region (Zemke et al., 2004). The only current effective therapy is thrombolysis, yet it is restricted to limited stroke patients. Neuroprotective agents with an extended therapeutic window that prevent multiple neurochemical cascades which ultimately cause irreversible brain damage are therefore urgently required.

Under physiological condition, the regulation of free radicals is controlled by endogenous antioxidant defense mechanism. Oxidative stress states the imbalance between the level of antioxidant defense mechanism and production of the free radicals. Many studies have clearly shown the involvement of oxidative stress in pathophysiology of neurodegenerative diseases (Loh et al., 2006a). During brain ischemia, reactive oxygen species (ROS) could alter the structural and functional integrity of cells by a variety of mechanisms, thereby causing severe cellular damage or even leading to cell death by necrosis or apoptosis. Of note, brain tissue is particularly susceptible to oxidative damage due to its high oxygen consumption, relatively low antioxidant level and low regenerative capacity (Kevin et al., 2004).

Intense interest has focused on the therapeutic effects of natural product on neurodegenerative diseases. In particular, natural product acts as antioxidant by enhancing the activity of endogenous antioxidants, preventing the free radical generation and neutralizing free radicals by nonenzymatic mechanisms (Zhu et al., 2004). We have previously shown therapeutic effects of one commercially available Chinese herb, *Ginkgo biloba*, on permanent middle cerebral artery occluded rats by influencing the apoptosis cascade in infarct area (Loh et al., 2006b).

In terms of Chinese medicine, *Herba Leonuri* (HL), also named Chinese Motherwort, it is prescribed during menstruation and child delivery in gynaecology. It acts to promote blood circulation by removing blood stasis when entering the blood system (Dan, 1993; Zhu et al., 1998; Zhu and Zhu, 2002). HL-based pharmaceutical preparations have also been used in China for the treatment of blood hyperviscosity and myocardial ischemia (Chen and Kwan, 2001) as it showed cardioprotective effect in previous studies (Zou et al.,

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1989; Pang et al., 2001). Most importantly, Liu et al. reported HL also has antioxidant properties by its strong superoxide-scavenging ability measured by an electron spin resonance (ESR) spin trapping technique in vivo (Liu et al., 2001).

Recently, raw HL was scientifically purified (Sun et al., 2005) and contains mainly stachydrine ($C_7H_{13}NO_2$), quercetin ($C_{15}H_{10}O_7$), Kaempferol ($C_{15}H_{10}O_6$), leonurine ($C_{14}H_{21}N_3O_5$) and apigenin ($C_{15}H_{10}O_5$). Our previous study showed that purified HL (pHL) preserves antioxidant properties and does have cardioprotective effect on ischemic myocardium through antioxidant effect (Sun et al., 2005). Currently, pHL is commercially available (Kardigen) as a potent heart supplement that protects the heart and reduces the risk of cardiovascular diseases.

In the present study, the therapeutic effect of pHL on stroke induced by permanent middle cerebral artery occlusion was evaluated. Our studies focused on the antioxidant effect of pHL and its ability on the reduction of apoptosis on rats subjected to permanent middle cerebral artery occlusion. Infarct volume and neurological deficit in each animal were targeted. Total antioxidant concentration and DNA oxidative damage were measured. In addition, TUNEL assay and immunohistochemical staining of various apoptotic related proteins were also carried out. We hope to clarify the underlying biochemical mechanisms of the antioxidant effect of pHL.

2. Materials and methods

2.1. Purified Herba Leonuri (pHL)

The raw material of HL material is originated from Sichuan province (China). Purified HL (pHL) is commercially available in Singapore (known as 'Kardigen', Herbatis Pte. Ltd., Singapore). pHL powder was supplied by Herbatis Pte. Ltd. (Singapore). The extract contains mainly stachydrine ($C_7H_{13}NO_2$), quercetin ($C_{15}H_{10}O_7$), Kaempferol ($C_{15}H_{10}O_6$), leonurine ($C_{14}H_{21}N_3O_5$) and apigenin ($C_{15}H_{10}O_5$). The preparation procedures of pHL were previously described in our previous study (Sun et al., 2005). Briefly, the raw material of HL was extracted by boiling in water for 45 min, and the aqueous extract was concentrated under vacuum at 50 8C (BUCHI Rotavapor R-144, waterbath B-480). After condensation overnight at 80 °C, the liquid was made into freeze-dried powder. The extract was analyzed by LC (liquid chromatograph)–ESI (electrospray ionization)–MS (mass spectrometry) (API 365 LC-MS system, Applied Biosystems, USA) (Sun et al., 2005). The chemical structures of these compounds are shown in Fig. 1.

2.2. In vivo experiments

The experimental protocol was approved by the ethical committee and confirmed to internationally accept ethical standards. The animals were supplied by Laboratory Animal Centre, National University of Singapore.

A total of 84 *Wistar* rats weighing 220–250 g were housed under diurnal lighting condition and allowed food and water ad libitum. All the animals were randomly divided into four groups: sham group with water treatment [Sham], sham group with pHL treatment (400 mg/kg/day) [Sham + pHL], stroke group with water treatment [Vehicle] and stroke group with pHL treatment (400 mg/kg/day) [Stroke + pHL]. The drugs were administered orally once daily. After 2 weeks of pre-surgery treatment, stroke was induced in the rats by middle cerebral artery occlusion (MCAO). In brief, rats were anaesthetized with ketamine/xylazine mixture (0.1 ml/100 g *i.p.*). Then, the middle cerebral artery (MCA) was exposed and occluded by electro-cauterization from the point where it crosses the inferior cerebral vein to a point proximal to the origin of the lenticulo-striate branch. The branches of the MCA between these two points were occluded as well to ensure complete and permanent occlusion. The wound was closed with suture. Rectal temperature was maintained at 37 ± 0.5 °C by means of a heating blanket throughout the surgery as reported previously (Lu et al., 2004). Sham rats undergone similar procedure with the omission of MCAO. Treatment was continued for another 1 week after surgery and the animals were sacrificed for sample collection.

2.3. Infarct volume measurement

The infarct volume was assessed with TTC (2,3,5-triphenyltetrazolium chloride) staining. After the brains had

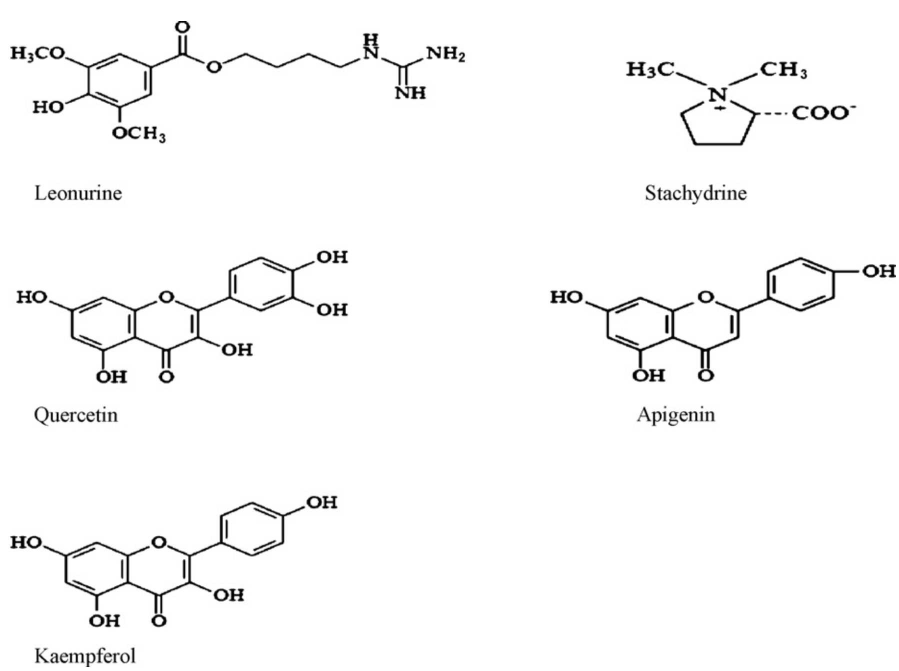

Fig. 1. Chemical structures of leonurine, stachydrine, quercetin, apigenin and kaempferol.

been collected, cerebellum and overlying membranes were removed. The brains were then sectioned into eight pieces of 2 mm thick coronal slices using a brain-sectioning block. The coronal sections were stained with 0.1% TTC solution at 37 °C for 30 min and preserved in 4% formalin solution. The infarct size was analyzed with an image analyzer system (Scion image for windows) and converted by integration (including correction for edema and atrophy) to the true infarct size of ischemic damage in the whole hemisphere as previously described (Lu et al., 2004).

2.4. Evaluation of neurological deficit

To evaluate neuronal function impairment after stroke insult, neurological deficit grading system was carried out as we described previously (Lu et al., 2005). A scale of zero to five was used to assess the behavioural and motor changes observed in the animal after the MCAO procedure. Rats were scored zero when they were suspended from the tail, they exhibited normal behaviour that includes extension of both forelimbs toward the floor. For those rats with contralateral forelimb to the side of the MCAO consistently flexed during the suspension and there was no other abnormality observed, they will be assigned a score of one. The rats were then placed on the ground and allowed to move freely and were observed for circling behaviour. Rats that moved spontaneously in all directions but showed a monodirectional circling toward the paretic side when given a light jerk of the tail were assigned a score of two. A score of three was assigned to rats that showed a consistent spontaneous contralateral circling. Rats that were very weak and walked only when stimulated were scored as four. If the rats died at the day of assessment, they were assigned a score of five. Rats that showed a higher clinical score exhibited all features of the lower grades. Therefore, rats with higher scoring showed more severe neuronal function impairment.

2.5. Total antioxidant assay

Plasma from each animal was collected for total antioxidant assay. A Total Antioxidant Status Assay Kit (*615700*, Calbiochem, Germany) was selected for the investigation of changes of endogenous antioxidant system under the influence of pHL. The assay was conducted according to manufacturer's instruction. The kit depends on the antioxidants in the sample inhibiting the oxidation of ABTS^{TM} (2,2′-azino-di-[3-ethylbenz-thiazoline sulphonate]) substrate to ABTS$^{TM\cdot+}$ product by a peroxidase, metmyoglobin. The amount of ABTS$^{TM\cdot+}$can be monitored by reading the absorbance at 600 nm.

2.6. DNA oxidative damage analysis using GC/MS

DNA was extracted from blood samples collected from each treatment group using the phenol–chloroform method. The DNA samples were then analyzed by GC/MS (Agilent 6890 gas chromatograph and interfaced with an Agilent MSD 5973) as previously described (Whiteman et al., 2002). DNA oxidative adducts analyzed by GC/MS from each sample were summed up and the mean value of DNA adducts was obtained.

2.7. TUNEL assay

At the end of treatment, the animals were perfused and fixed with 2% paraformaldehyde. The brains were then collected. Brains collected were post-fixed in 2% paraformaldehyde for 2 h. They were then transferred into 15% sucrose in phosphate buffer overnight and subsequently 30% sucrose in phosphate buffer to allow the sample dehydration. Brain tissues were cut at 20 μm using a Leica® CM1510 Cryostat (Leica Microsystem, IL, USA). A DeadEnd™ Fluorometric TUNEL System (*G3250*, Promega, USA) kit was chosen for detection of apoptosis-induced nuclear DNA fragmentation via fluorescence microscopy. Slides were viewed and photographed using a fluorescent Leica® microscope (Leica Microsystems, IL, USA) using standard fluorescein filter set to view the green fluorescence of fluorescein at 520 ± 20 nm while blue DAPI at 460 nm.

2.8. Immunohistochemical staining

In addition, immunohistochemical staining (IHC) was carried out using frozen sections. A ready to use UltraVision Detection System Anti-Polyvalent, HRP/DAB kit (Lab Vision Corporation, CA, USA) was used for antibody staining. Polyclonal rabbit anti-Bax antibody (*sc-493*, Santa Cruz, CA, USA), polyclonal rabbit anti-Fas antibody (*sc-715*, Santa Cruz, CA, USA) and polyclonal rabbit anti-Bcl-xL antibody (*sc-492*, Santa Cruz, CA, USA) and Bcl-2 (*sc-7195*, Santa Cruz, CA, USA) were used for the primary antibody incubation. Hematoxylin staining was selected as counterstaining. Slides were viewed and photographed using a fluorescent Leica® microscope.

2.9. Statistical analysis

Data was represented as mean ± S.E.M. Statistical analysis was performed by the two-tailed Student's *t*-test for difference between samples. A difference with $p < 0.05$ was considered statistically significant.

3. Results

3.1. Infarct volume measurement

Infarct volume of each treatment group is shown in Fig. 2. As expected, the rats from sham and sham treated with pHL

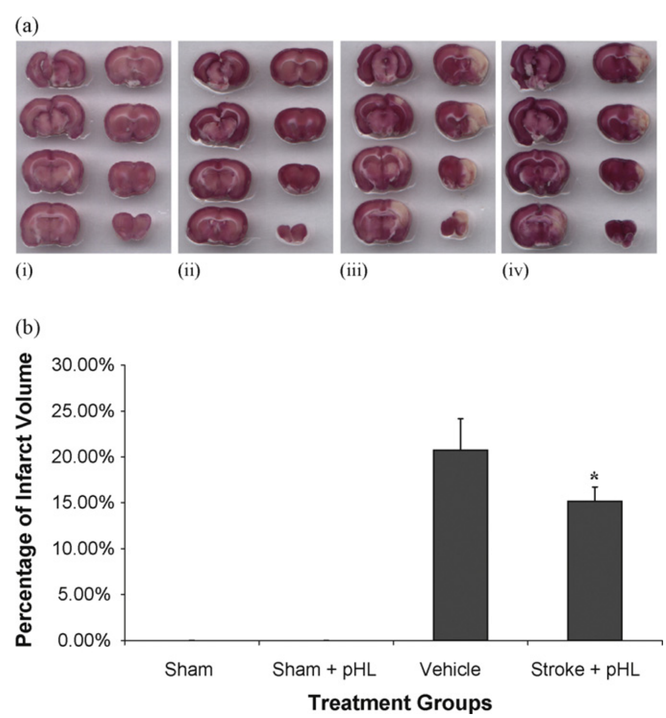

Fig. 2. (a) Infarct area of each treatment group was observed by TTC staining. (i) Sham; (ii) Sham + pHL; (iii) vehicle; (iv) stroke + pHL. (b) Infarct volume was then analyzed by image analyzer system (Scion image for windows). Under the treatment of pHL, infarct volume was greatly reduced as compared to vehicle group. $^{*}p < 0.05$ pHL treated-stroke operated vs. vehicle treated group.

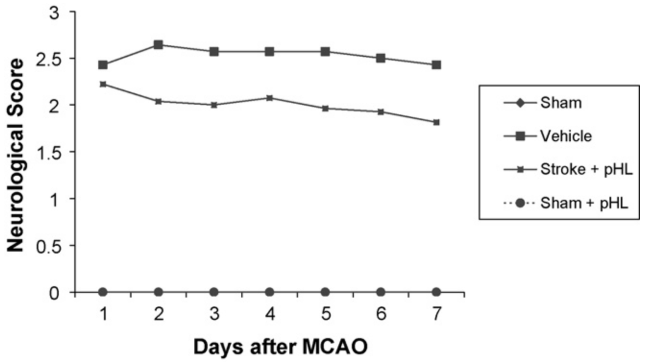

Fig. 3. Neurological deficit score among treatment groups. The higher the scoring, the more severe the motor impairment is. Vehicle group showed the highest neurological deficit score while HL stroke group showed a significant reduction of neurological score at the end of treatment (day 7).

had no cerebral injury, therefore no infarct area was observed (Fig. 2ai and ii). When the animal was subjected to ischemia insult by left MCAO, the infarct area was observed in left cortex and striatum (Fig. 2aiii). Under the treatment of pHL, the infarct volume was reduced significantly from 20.75 ± 0.03% to 15.19 ± 0.02% (Fig. 2b).

3.2. Neurological deficit grading system

In terms of evaluation of neurological function, neurological deficit grading system was carried out for all the animals. The higher the neurological deficit score, the more severe impairment of motor motion. The result is shown in Fig. 3. As rats from sham and sham treated with pHL group undergone sham operated surgery, they did not have any neurological deficit, and therefore throughout the entire study, the animals from both groups had the neurological score of zero. For the rats in vehicle group, they remained highest neurological deficit score after the surgery. This is in agreement with that they had largest infarct volume among the groups (Figs. 2 and 3). Under the treatment of pHL, neurological deficit score of rats undergone MCAO was lowered as compared to vehicle group (2.22 vs. 2.43) at day 1. As the treatment continued, their neurological deficit score declined from 2.22 to 1.81 (Fig. 3).

3.3. Total antioxidant status assay

To investigate the changes of endogenous antioxidant system under the influence of pHL, total antioxidant assay was carried using plasma sample. The kit depends on the antioxidants in the sample inhibiting the oxidation of ABTS™ substrate to $ABTS^{TM\cdot+}$ product by metmyoglobin. Therefore, the concentration of antioxidant present in the sample is inversely proportional to the amount of $ABTS^{TM\cdot+}$ to be measured. The concentration of antioxidant of sham group was 0.38 ± 0.08 mM (Fig. 4a). Under the treatment of pHL, the antioxidant concentration was increased from 0.38 ± 0.08 mM to 0.5 ± 0.2 mM though it is not statistically significant (Fig. 4a). This might be due to the sensitivity of the system which requires a very accurate time point measurement. As for the vehicle group, the antioxidant concentration was decreased significantly to 0.31 ± 0.03 mM as compared to sham group (Fig. 4a). Under pHL treatment, the antioxidant concentration was restored to 0.42 ± 0.05 mM, level that is even higher than sham group (Fig. 4a).

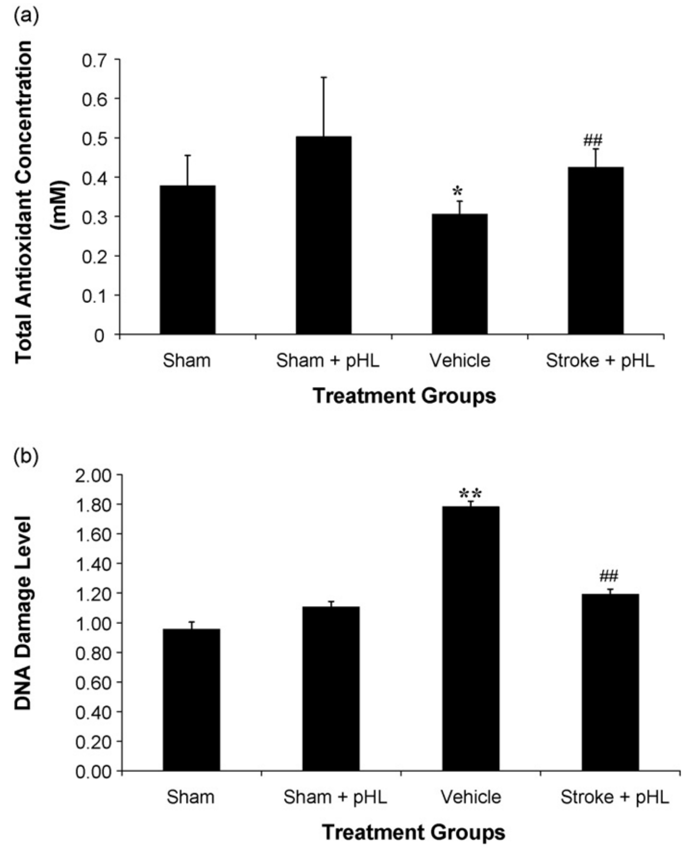

Fig. 4. The antioxidant profile of each treatment group under the influence of the pHL. (a) The concentration (mM) of total antioxidant present in plasma among the treatment groups. Total antioxidant concentration was enhanced in both sham operated and stroke operated groups. $^{*}p < 0.05$ vehicle treated group vs. sham treated group. $^{\#\#}p < 0.01$ pHL treated-stroke operated vs. vehicle treated group. (b) The DNA oxidative damage level in each treatment group. Under the treatment of pHL, DNA oxidative damage was reduced significantly in stroke-operated group while pHL did not alter the oxidative damage level in healthy individual. $^{**}p < 0.01$ vehicle treated group vs. sham treated group. $^{\#\#}p < 0.01$ pHL treated-stroke operated vs. vehicle treated group.

3.4. DNA oxidative damage analysis

DNA oxidative adducts could be considered as one of the marker of oxidative stress. To measure the level of oxidative stress in each treatment group, DNA oxidative damage analysis was carried out using GC/MS. As shown in Fig. 4b, pHL treatment did not alter the basal level of DNA oxidative damage as the level of DNA oxidative damage is almost the same in both sham group and sham treated with pHL group, 0.98 ± 0.05 and 1.08 ± 0.06, respectively. Vehicle group has the highest level of DNA oxidative damage among all the groups (1.78 ± 0.03) as shown in Fig. 4b. pHL had a significant effect on reducing the oxidative damage level as reduction of the DNA oxidative damage to 1.19 ± 0.03 in stroke operated pHL treated group was observed (Fig. 4b), which was almost comparable to sham group.

3.5. TUNEL assay

The apoptotic cells were identified through TUNEL staining. In TUNEL staining, apoptotic cells exhibited strong, nuclear green fluorescence. Apoptosis was detected in the infarct area of left cerebral cortex, while no apoptosis was detected in non-infarct area, sham-operated rats (Fig. 5a) and sham-operated pHL treated rats (Fig. 5b). Strongest apoptosis marker was observed in vehicle

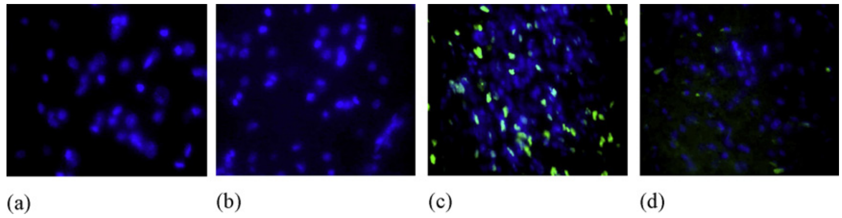

Fig. 5. Apoptotic staining in cerebral cortex after 7 days of MCAO (20× magnification) for each treatment group. Slides were viewed at 520 ± 20 nm to detect nuclear green fluorescence and viewed at 460 nm to detect DAPI staining. Images shown were combined (overlapped) to detect overall morphology of cell population. (a) Sham; (b) Sham rats treated with pHL; (c) stroke rats; (d) stroke rats treated with pHL.

group (Fig. 5c) which is correlated with the largest infarct volume and most severe neurological impairment. pHL treated group had significant reduction of apoptosis marker in the infarct area (Fig. 5d).

3.6. Immunohistochemical staining (IHC)

The localization of the apoptotic related proteins was identified by IHC. Negative controls were included to ensure the specificity of secondary antibody (Fig. 6a). No pro-apoptotic protein BAX staining was observed in both sham group and sham-operated pHL treated group (Fig. 6b). In contrast, strongest signal of BAX was observed in vehicle group (Fig. 6b). Weaker BAX immunoreactivity was observed in stroke operated pHL treated group (Fig. 6b). These positive staining of BAX was observed merely in infarct area. Similar results were observed in each treatment group for FAS, as shown in Fig. 6c. Pro-apoptotic proteins BAX and FAS had been dramatically increased in the infarct area after MCAO as seen in the vehicle group. Furthermore, these pro-apoptotic proteins could only be detected in infarct area, indicating the up-regulation of these proteins in infarct area during ischemia as compared to undetectable levels of these proteins in non-infarct area.

For both anti-apoptotic proteins, BCL-2 and BCL-XL (Fig. 6d and e), strongest positive signals could be observed in both sham and sham-operated pHL treated group. A very weak BCL-2 positive staining and no BCL-XL signal were detected in vehicle group (Fig. 6). The positive signals of both BCL-2 and BCL-XL were restored in stroke operated pHL treated group as seen in Fig. 6.

4. Discussion

During ischemia, the disruption of cerebral blood flow deprives brain cells of oxygen, leading to a reduction in energy production associated with building-up toxic metabolites such as glutamate,

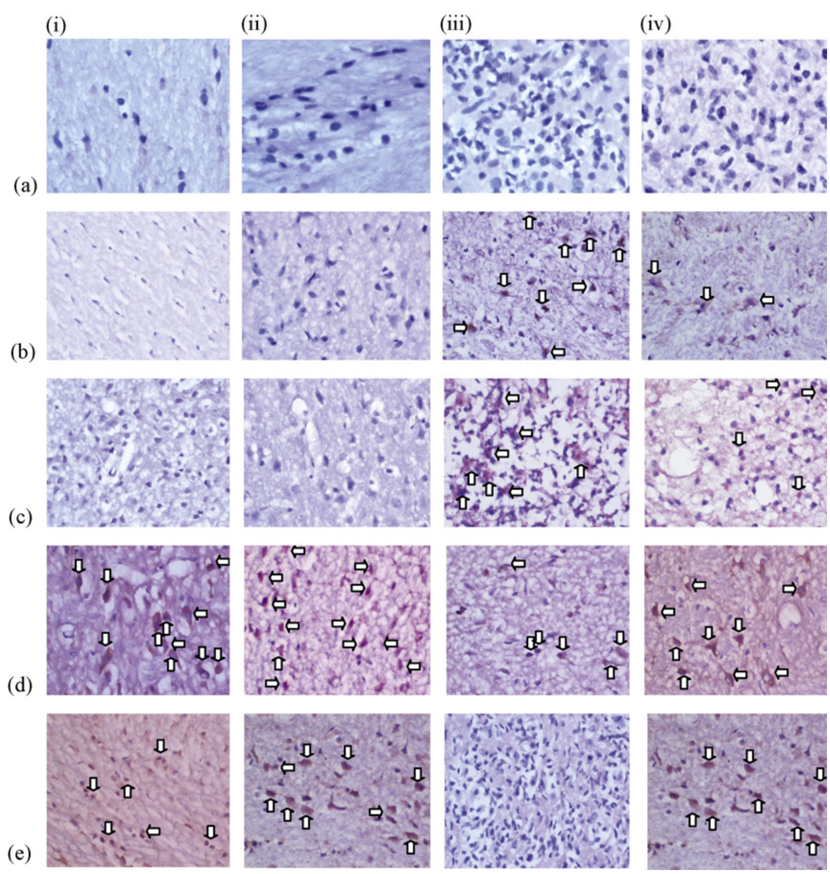

Fig. 6. Immunohistochemical staining of pro-apoptotic protein (b) BAX and (c) FAS and anti-apoptotic protein (d) BCL-2 and (e) BCL-XL in cerebral cortex after 7 days of MCAO (40× magnification). Negative control (a) was included. The staining showed stronger immunoreactivity of pro-apoptotic proteins and diminished immunoreactivity of anti-apoptotic proteins after stroke induction. Following the HL administration, enhanced immunoreactivity of anti-apoptotic proteins and weaker pro-apoptotic proteins immunoreactivity were shown. From left (i) Sham; (ii) Sham treated with pHL; (iii) vehicle; (iv) stroke rats treated with pHL.

inflammatory mediators, free radicals that could ultimately trigger ischemic cascade. As mentioned, over production of free radicals during ischemia results in oxidative stress. Several possible mechanisms have been postulated for the generation of ROS during ischemia. Firstly, in addition to the basal level of superoxide production by mitochondria during oxidative phosphorylation, reduced oxygen level causes the disruption of electron transport chain, therefore increases the leakage of electron and production of superoxide production. Increased arachidonic acid metabolism during ischemia produces free radicals as by-products. In addition, elevated Ca^{2+} might cause ROS production through xanthine metabolism. Furthermore, inflammatory response is also associated with free radical generation (Loh et al., 2006a). Oxidative stress does not play in isolation, but participates in the complex interplay between excitotoxicity, apoptosis, inflammation and so forth. Low antioxidant defence mechanism and high oxygen consumption subject the brain to be more vulnerable to the oxidative stress insult. Therefore, modulation of oxidative stress could be a potential therapeutic approach for various neurodegenerative diseases.

The results from this study demonstrate for the first time that pHL could have neuroprotective effects particularly in ischemic stroke. In this study, pHL could reduce the infarct volume and improve the neurological impairment caused by MCAO. In addition, it enhanced the endogenous antioxidant capacity and conferred a protective mechanism against ischemic insult, therefore, reduction of DNA oxidative damage and apoptosis were also resulted.

In order to resemble a massive and potentially fatal ischemic stroke in humans, MCAO is an extensively used model of focal cerebral ischemia. The infarct formed by MCAO is similar to the brain damage of ischemic stroke in humans (Miller, 1999). In the case of left middle cerebral artery, it supplies the blood to left cortical areas and also corpus striatum. Damage of these areas results in impairment of motor, speech and swallowing functions. In our study, we demonstrated left MCAO caused the infarction at the area in left cortical area and corpus striatum shown by TTC staining. Furthermore, our study also demonstrated that rats suffering from left MCAO had the severe impairment of motor function as measured by neurological deficit grading system. Under the treatment of pHL, the infarct volume was significantly reduced. This phenomenon indicates the lesser histological damage was observed as compared to those without treatment. Therefore, the neurological impairment was also alleviated under the treatment of pHL.

We believe the protective effect of pHL is largely due to its antioxidant property. Therefore, we investigated the endogenous antioxidant system under the influence of pHL. Plasma was selected as the sample for this experiment as it contains many nonenzymatic antioxidants and antioxidant enzymes. The major antioxidant defences in plasma include ascorbate, protein thiols, bilirubin, urate and α-tocopherol (Nicholas et al., 1993). Plasma contains also the preventive antioxidants, caeruloplasmin and transferrin, the iron-scavenging proteins which can contribute to the total antixodant capacity because it scavenges the iron availability (Stocks et al., 1974). Our study showed the total antioxidant concentration decreased significantly after MCAO. This is in consistent with the work done previously by Antonio et al. (2000), who showed that mean plasma levels of nonenzymatic antioxidants and antioxidant enzymes activities were lower in patients than in healthy controls (Antonio et al., 2000). Sun et al. (2005) also showed that rat undergone myocardial infarction had lower level of antioxidant enzyme activities in heart (Sun et al., 2005). The low level of antioxidant capacity in stroke groups indicates that much of the antioxidant molecules have been used up and react with ROS which was over produced during ischemia. Under the pHL treatment, both in healthy or stroke group, the plasma antioxidant concentration was greatly enhanced. This implies that pHL may have enhancing effects on endogenous antioxidants with or without oxidative stress condition, thereby confer protective barrier against stroke.

We also evaluated the level of DNA oxidative damage as one of the marker of oxidative stress. GCMS has of late been widely used to quantify DNA damage and the level of oxidative stress due to its ability to identify a wide range of DNA base product in a single analysis. For the DNA oxidative damage analysis, we focused on direct damage by ROS that can affect purine, pyrimidine bases and/or deoxyribose sugar (Halliwell and Gutteridge, 2007). From our study, it showed that DNA oxidative damage was increased after MCAO; suggesting the increased free radical activity, as well as reciprocal decreased endogenous antioxidant activity contribute to DNA oxidative damage after stroke. Rats with stroke insult treated with pHL not only had high endogenous antioxidant concentration but also had significant reduction on DNA oxidative damage. We therefore propose that the in vivo therapeutic effects of pHL might be strongly related to its antioxidant effects by enhancing the endogenous antioxidant capacity and also alleviating the oxidative stress.

In focal ischemia, there is rarely complete blockade of blood flow to the affected region because plethora of collateral vessels provides some flow to the area, making the affected region a core infarct area and an ischemic penumbra which receives collateral flow that is sufficient to prevent cell death by necrosis. The cells in this region maintain a very low metabolic rate, electrical silence. Depending on the severity of ischemia, if the ischemia is short lasting, the cells might recover from no apparent injury. If the ischemia is long-lasting, they might undergo delayed type programmed cell death as they are bombarded by waste products from the dead cells in infarct core (Suresh et al., 2007). Many studies have demonstrated that delayed cell death in ischemic penumbra is attributed to apoptosis and rational treatment of stroke has likewise focused on protecting this area of cells.

In past, direct treatment of oxidants like hydrogen peroxide was thought to induce exclusively necrosis. However, more recent studies have shown that lower doses of free radicals can trigger apoptosis as well. Reciprocally, broad-spectrum of anti-apoptotic proteins such as BCL-2 have been ascribed as an antioxidant function by maintaining cells in a more reduced state through scavenging ROS either directly or up-regulating other ROS scavengers such as thiol compounds and antioxidant enzymes, further indicating that ROS generation is one of the crucial initiation step for apoptotic event (Chandra et al., 2000). Recently, it has also been shown that oxidative stress can lead to apoptosis by inducing mitochondria permeability transition (MPT) pore opening. ROS can also lead to oxidation of the anionic phospholipids cardiolipin to facilitate cytochrome *c* release from mitochondria (Halliwell and Gutteridge, 2007).

Since ample evidences demonstrate the link between oxidative stress and apoptosis, we would like to investigate whether pHL could ameliorate the level of apoptosis during stroke. The TUNEL system is designed for the detection of nuclear DNA fragmentation, an important biochemical hallmark of apoptosis in many cell types. Results obtained from TUNEL assay showed that green fluorescence could only be detected in infarct zone, indicating that apoptosis occurred only at the affected region. In addition, strongest nuclear green fluorescence was observed in stroke-induced rats, which is correlated with the results obtained from other analysis. Lesser nuclear green fluorescence was observed in the stroke rats treated with pHL. Together with its ability of reducing infarct volume, neurological impairment and DNA oxidative damage, result obtained from TUNEL assay further emphasizes the therapeutic potential of pHL in the middle cerebral artery occluded *Wistar* rats.

Generally, apoptosis can be classified as intrinsic pathway and extrinsic pathway. There are considerable cross-talks between both pathways. Subsequently, they converge to result ultimately in cellular morphological and biochemical alterations (Ashe and Berry, 2003). Central to intrinsic pathway are proteins in the Bcl-2 family. The Bcl-2 family can be subdivided into pro- and anti-apoptotic proteins. Bax belongs to the pro-apoptotic protein while Bcl-XL and Bcl-2 belong to anti-apoptotic protein (Love, 2003). For Fas, it is involved in the extrinsic pathway of apoptosis (Ashe and Berry, 2003). Our immunohistochemical staining results showed the presence of pro-apoptotic proteins BAX and FAS only in the infarct area after MCAO but undetectable in the healthy individuals. Highest immunoreactivity of pro-apoptotic proteins in vehicle group is correlated with observation of strongest nuclear green fluorescence in vehicle. This indicates the up-regulation of pro-apoptotic proteins and thereby intense cell death with dramatic morphological changes occurred in the region of infarct area. In contrast, anti-apoptotic proteins BCL-XL and BCL-2 were found in entire brain section in both sham and sham with pHL group. Weaker signal was observed in vehicle group. Significant reduction of immunoreactivity of pro-apoptotic proteins and enhancement of anti-apoptotic protein signals were observed in stroke rats with drug treatments. This observation leads us to conclude that pHL with antioxidant property could have effect on apoptosis by switching the cells toward more anti-apoptotic state.

Although pHL has shown its therapeutic effect against stroke in vivo, the underlying mechanisms and the feasibility of long term usage are yet to be verified. Therefore, a more detailed experiment could be conducted for the understanding of mechanisms of pHL particularly on its antioxidant properties. Future studies can also be done on the purification of single compound from pHL to identify the active compounds of pHL.

5. Conclusion

Oxidative stress process is a complex interplay that involves in cellular damage and cell death. Therefore, studies focusing on antioxidant properties of Chinese herbs have been shown its promising therapeutic effects against stroke. In this study, pHL has shown its therapeutic potential on reducing infarct volume, neurological impairment, oxidative damage and apoptosis, probably through its antioxidant effect, at least by the ability of increasing the endogenous antioxidant capacity. Therefore, pharmacological modification of oxidative damage is one of the most promising avenues.

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